THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the number 4721, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 6084 **(b)** 6264 **(c)** 6174 **(d)** 6183

2. The sum of the three consecutive numbers 25, 26 and 27 is: 1

(a) 75 **(b)** 76 **(c)** 81 **(d)** 78

3. The sum of the five consecutive numbers starting from 29 is: 1

(a) 155 **(b)** 145 **(c)** 150 **(d)** 160

4. Think of 9. Add 4, multiply by 3 and subtract 12. The result is: 1

(a) 31 **(b)** 23 **(c)** 12 **(d)** 27

5. The smallest digit that can replace * in 5*34 so that the number is divisible by 3 is: 1

(a) 4 **(b)** 1 **(c)** 0 **(d)** 7

6. The product of three consecutive natural numbers is always: 1

(a) a prime number **(b)** odd **(c)** even **(d)** a perfect square

7. Is 5217 divisible by 11? 1

(a) Only if the number is a perfect square **(b)** No

(c) Yes **(d)** Only if the last digit is even

8. Is 6150 divisible by 6? 1

(a) Yes **(b)** No

(c) Only if the last digit is even **(d)** Only if the number is a perfect square

9. For the number 9053, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 9180 **(b)** 9081 **(c)** 9261 **(d)** 9171

10. The sum of the three consecutive numbers 37, 38 and 39 is: 1

(a) 117 **(b)** 112 **(c)** 114 **(d)** 111

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Apply one step of the Kaprekar routine to 8741. 2

12. Take 21. Add 4, multiply by 5, then subtract 20. What is the result? How is it related to 21? 2

13. Use the divisibility rule for 11 to test 9152. 2

14. Apply one step of the Kaprekar routine to 2091. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Check whether 49896 is divisible by 8 and by 11. 3

16. Apply the Kaprekar routine to 2963 repeatedly until you reach 6174. Write each step. 3

17. Show that the sum of 29 and its reverse is divisible by 11. 3

OR

Check whether 91728 is divisible by 8 and by 11. 3

18. Apply the Kaprekar routine to 5108 repeatedly until you reach 6174. Write each step. 3

OR

Show that the sum of 85 and its reverse is divisible by 11. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Replace the stars so that the number 2*4* is divisible by both 10 and 9. Find all possibilities. 5

OR

Take 94 and its reverse 49. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

20. Replace the stars so that the number 8*1* is divisible by both 5 and 9. Find all possibilities. 5

OR

Take 52 and its reverse 25. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

END OF QUESTION PAPER — Total: 40 Marks

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